

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		award (1) for each correct formula A H ₂ B HCl C PbCl ₂		3		3		
		(ii)		C	1			1		1
	(b)	(i)		neutralisation	1			1		
		(ii)	I	12.4		1		1		1
			II	19.0		1		1		1
			III	the acid and the alkali have the same concentration ☐ the acid has a lower concentration than the alkali ☐ the concentration of the acid is half of the concentration of the alkali ☐ the acid has a higher concentration than the alkali ☒			1	1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		award (1) for any of following • Eleanor knew the exact pH (at any point) • pH sensor gives an accurate / precise pH (at any point) • Freddie only knew an approximate pH value (at any point) • universal indicator only gives an approximate pH value • universal indicator does not give accurate / precise pH values • universal indicator does not give a colour for pH 2.5 / 4.5 / 6.5 etc • not possible to judge when one colour becomes another e.g. when orange becomes red accept any sensible answer neutral answers no room for human error more reliable			1	1		1
				Question 5 total	2	5	2	9	0	5